Dear Professor Seim,

Thank you for granting me the opportunity to re-submit my paper "Regulating Competing Payment Networks" to the American Economic Review.

I deeply appreciate your comments on how to revise the paper in order to maximize its potential. I am also thankful to the four referees, who carefully read the previous version and provided me with a clear path for this revision. I think that the paper has improved substantially thanks to the suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.

2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss how I modified the paper and extended my analysis to address it.

I also discuss all points raised by the referees and, wherever possible, address them empirically. I have attached four separate letters with my detailed answers to each referee.

Thank you again for your comments and clear guidance in revising the manuscript.

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification. This responds to concerns by R1 about the interpretation of the merchant event studies, by R1 and R2 on the Discover evidence, and your concerns about the Durbin evidence.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed. This responds to your guidance on data limitations, R2's comments on fixed costs, and R4's point about short-run predictions.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix OF estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit. This responds to your concerns and those of R1–R3 about credit-debit substitution.

4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results ($31 bn vs. $27 bn). This responds to R1's suggestion to decompose welfare effects and investigate multi-homing, R2's suggestion for a moderate fee cap, R4's suggestion on the social

optimum, your suggestion on evaluating robustness to reward sensitivity, and your request to check whether credit constraints affect the results.

5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout. This responds to your request and that of the AE to tighten the exposition.

Detailed Response to the Editor Letter

> (a) Caveat the results in Section III.A given the sensitivity to sample selection. (b) Address Referee 4 concerns that the difference-in-difference results are unable to isolate the role of rewards and that the estimated rewards sensitivity is too high. You already note that the control group aims to purge the results of some types of contemporaneous trends, such as changes in merchant acceptance behavior. Can you provide any evidence on potential strategic responses by exempt issuers to the Amendment, such as promotion of debit cards? Alternatively, can you - as a robustness check - reduce the sensitivity to rewards in model estimation?

Reply: I address these concerns with a reframing of the reduced-form evidence around multiple steering channels, new bank-level data on the rewards channel, a note on the welfare implication of non-rewards steering, and a robustness check that halves the targeted Durbin moment.

Reframing the reduced-form evidence. Section III.A now presents the Durbin evidence as showing that interchange fees steer consumer payment choices through multiple channels. The Durbin Amendment cut large issuers' interchange revenue, and those issuers responded across several margins: cutting rewards, reducing advertising, and scaling back teller incentives. Their debit volumes fell. The main text cites qualitative industry evidence supporting all three channels.

The main text also now notes that the decline is weaker once the full sample of debit card payment volumes is included.

New data on the rewards channel. I collected bank-level data from archived bank websites on which issuers paid debit rewards before and after Durbin. Among the fifteen banks that paid debit rewards pre-Durbin, those that ended rewards saw a decline in debit volumes of about 25% relative to those who did not cut rewards, matching the baseline difference-in-difference estimate. I refer to this analysis in the main text in Section III.A.

In Section V.A.1, I connect this 25% to the rewards channel specifically. Banks fund debit promotion from interchange revenue, allocating across rewards, advertising, and teller incentives. If banks differ in their productivity across these levers, rewards-paying banks substituted rewards for non-price promotion and did little non-price steering to begin with. The within-subsample comparison therefore reflects mainly the rewards change. Thus the baseline estimates do not overstate the role of rewards.

Robustness check. In Section VI.F, I now re-estimate the model while halving the targeted Durbin moment, so the model matches only half the observed decline in debit volumes. The estimated reward sensitivity drops by around half (the log price sensitivity coefficient declines by around 0.83). Halving the moment produces negative estimated marginal costs for Visa, which are difficult to reconcile with accounting data. The qualitative welfare conclusions for the fee cap, Durbin, CBDC, and dual-routing counterfactuals are nonetheless preserved. The sign of the merger counterfactual flips, however, as networks have more market power over consumers.

In the same section, I also emphasize that if non-rewards steering (advertising, teller incentives) imposes real costs on banks without generating direct consumer benefits, the baseline welfare estimates *understate* the true losses from high merchant fees. Lowering merchant fees would also reduce steering, which would raise total welfare.

> (a) Please summarize retailer credit card acceptance rates upfront - my sense is that this grocery store must be relatively unique in not accepting credit cards. This will help motivate modeling assumptions. (b) Clarify the interpretation of the results. I agree with the R1 that one takeaway of the grocery store analysis is that debit and credit cards substitute and that the Discover evidence alone is not sufficiently conclusive to rule out this channel fully.

Reply: I now clarify early on in Section III.B that the merchants I study are highly selected: 98% of trips in Homescan occur at stores that accept credit cards. This helps motivate the estimation strategy that treats the estimated sales effects as the gains for marginal merchants who decide to accept credit cards rather than the average merchant.

I have narrowed the interpretation of my event study results in Section III.B around two main points.

1. The event study results show that incremental acceptance of credit cards increases sales for merchants, even though most consumers with credit cards own debit cards.

2. The Discover evidence suggests that consumers do not change their consumption decision between stores in response to rewards.

I agree that the Discover evidence alone is not sufficiently conclusive to rule out debit-credit substitution. I defer the discussion of the substitution assumption to the model, where I estimate two alternative specifications that navigate the trade-off between consumer-side evidence on substitution and merchant-side evidence on acceptance and pricing differently (Appendix OF).

As an aside, my understanding of the Nielsen guidelines is that it is not possible to display results based on data from a single entity. Please ensure that the paper passes review by the Kilts Center.

Reply: The event study contains two grocery chains. The paper has passed Kilts review.

At a high level, the paper needs to be more open about the fact that the merchant data and the current state of merchant card adoption do not allow you to model merchant behavior as flexibly as one may like. A few examples:

(a) You rely on the single, possibly not representative merchant from Section III.B to pin down σ and use this single retailer to support the empirical result that merchants are fee-insensitive (presumably, to current levels of fees!). This is more like a calibration than an estimation exercise, however, that likely plays a role in the overall model predictions.

(b) The fact that most merchants accept some type of credit card today and that you have already exploited Section-III.B grocery store's adoption behavior in estimation must mean that the data do not allow you to identify fixed costs to card acceptance separately from merchant substitution and heterogeneity. Please correct me if this is wrong. If not, you should simply state this as a shortcoming of your approach and explain why you cannot be more flexible, rather than argue that fixed adoption costs are small (which R2 does not agree with). Please note that I do not expect you to incorporate fixed acceptance costs, but please add a discussion of how such costs might affect agent behavior; R2's report contains nice conjectures on this point.

(c) You say that you abstract from non-rewards characteristics because of Australia's experience of regulating merchant fees (Section IV.F.7). I assume, however, that your data does not contain information on non-rewards characteristics of consumers' credit cards and that modeling issuer choices in dimensions other than fee structures is well beyond the scope of the paper. As R4 notes, the counterfactuals are therefore best thought of as short-run predictions, with possible regulations inducing further strategic responses by issuers. R2's suggestion of representing such concerns as an artificial quality degradation in the counterfactuals is nice, but I do not necessarily expect you to implement it.

Reply: U.S. payment markets are mature, limiting the variation available for studying merchant behavior. The grocery chains that changed acceptance policies are the only natural experiments in the data. I have revised the paper to state data limitations directly wherever they constrain the analysis, rather than defending modeling choices as first-best.

The merchant-type estimation (Section V.A.3) is now framed as a calibration exercise anchored by the grocery-chain event studies, with an explanation of why more flexible specifications are not feasible given the data. In Section IV.F, I state that fixed costs of card acceptance cannot be identified separately from heterogeneity in sales benefits, and discuss how they might affect behavior following R2's conjectures. Following R4's concerns, the counterfactuals are now described as short-run predictions that hold non-rewards characteristics fixed. The same approach extends to other limitations: credit-debit segmentation and one-dimensional merchant heterogeneity are discussed in Section IV.F, and the pass-through specification in Section VI.F.

> The referees continue to ask for improvements in your treatment of credit and debit cards on the consumer side. First, R1-R3 dislike that they do not substitute. They also make several suggestions for how you might introduce substitution in the model (see R1 and R2's comments here) or clarify why exactly you make this assumption (see R3's suggestion on this point). Since your consumer-side data are better, I wondered if you could make some progress on this point, but I would also be satisfied with a more detailed motivation for the need for this assumption based on modeling challenges or data shortcomings that this might introduce.

Reply: While incorporating credit-debit substitution makes the model more realistic along some dimensions, it also generates other model predictions that are at odds with the empirical evidence. As such, the main text keeps the assumption of point-of-sale segmentation between credit and debit.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix OF.1). Alaska Airlines, Best Buy, and Crate & Barrel describe credit and debit as "totally different products that cannot substitute for each other." On the other hand, Discover's rewards shock (Online Appendix OD.2) shifts consumer spending onto Discover from both cash and debit when Discover pays higher grocery rewards, consistent with some substitution at the point of sale.

The main challenge with modeling credit-debit substitution is that it generates two equilibrium predictions that are at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates, but expert witnesses in *Ohio v. AmEx* instead describe merchants disciplining credit fees by threatening to route transactions to rival credit cards rather than debit. The second is that capping debit merchant fees would lower credit merchant fees, but the Durbin Amendment cut regulated debit interchange by roughly half without moving credit interchange (Figure 5).

The segmentation assumption in the main text (Section IV.F) prioritizes matching the macro facts on merchant acceptance and pricing, at the cost of imperfect fidelity to the micro evidence on consumer substitution. Online Appendix OF estimates two specifications that navigate the trade-off between the macro and micro facts differently. The first allows point-of-sale substitution between credit and debit and incremental debit sales relative to cash (Online Appendix OF.3.1). This matches the micro evidence most closely. The second treats debit as generating no incremental sales relative to cash (Online Appendix OF.3.2). This gives up some fidelity to the micro evidence, but it matches the fact that merchants do not incorporate credit-debit multi-homing into their credit card acceptance decisions. It is silent on Durbin pricing because debit fees are zero by construction.

The welfare effect of repealing the Durbin Amendment flips sign across the specifications that can evaluate it, since the assumption determines how much credit networks would cut credit merchant fees in response. Price and welfare effects for the credit fee cap, merger, and dual routing counterfactuals are quantitatively similar across specifications.

> Second, R4 notes that not all consumers are able to obtain credit cards. Translating this to the model means that your choice set is mis-specified for a segment of consumers. My conjecture was that with your current setup, such constraints would lead to an understatement of the relative value of credit cards. Please clarify whether this is correct. Are you able to perform a crude robustness check by re-estimating the model under the assumption that a segment of (lower income) consumers do not have access to credit cards?

Reply: Your conjecture is correct: ignoring credit constraints understates the relative value of credit cards. However, this does not change the estimated welfare results.

Section VI.F discusses this alternative interpretation in the main text, and Online Appendix OG.2 re-estimates the model under the assumption that credit card access varies with income, using DCPC data on credit scores to parameterize this relationship. As expected, the median consumer's credit aversion is smaller because we no longer need to rationalize the low observed credit card usage through preferences. The income gradient in credit preference also weakens, since income-dependent access now absorbs much of the correlation between income and credit card use.

The welfare results are similar ($27 bn in the baseline and $32 bn in the credit-constrained specification). Both models are estimated to match the same Durbin Amendment moment (the observed shift in payment shares after the debit fee reduction). Constrained consumers respond less elastically on the credit margin, so the remaining unconstrained consumers must be more reward-sensitive to rationalize the same aggregate response.

The predicted change in shares from a fee cap is therefore similar, and the policy ranking is unchanged across all counterfactuals.

> Third, R2 suggests clarifying the information environment for consumers, its interaction with the consumer merchant choice, and the role of card preferences in merchant choice.

Reply: I have clarified the information environment in Section IV.B.2. Consumers observe merchants' acceptance decisions and prices before choosing where to shop. In Section IV.C.3, I also clarify that consumers have complete information over which merchants accept cards when they choose which cards to adopt.

> I agree with R2 that reducing merchant fees down to the cost of capital is unlikely (point 2c and point 2a, even though I do not expect you to simulate a counterfactual where credit cards are eliminated from the market, which is interesting, but maybe not as relevant in the current environment). The referee therefore suggests a more nuanced reduction sequence that would also illustrate the key features of the model better, as requested by R4.

Reply: The main fee cap counterfactual is now a more moderate 120 bps cap, matching Canadian merchant fees. By changing the fee cap, I no longer need to extrapolate as far from the current equilibrium, which helps to address concerns from R2 about fixed costs.

Online Appendix OG.4 compares the moderate 120 bps cap to the cost-of-cash cap and the Ramsey planner's solution. The relationship between the cap level and welfare is concave: even moderate caps generate large benefits. The cost-of-cash cap delivers only 7.4% higher welfare gains than the moderate cap that is half-way between current fee levels and the cost of cash.

> In the spirit of better illustrating model behavior, I think R1's suggestion of decomposing the fee cap effect into different channels would be very useful.

Reply: I have adopted R1's suggestion and decompose welfare effects by sequentially allowing more merchant responses. Table 6 in Section VI.A reports this decomposition for the 120 bps cap. The first row (Network Effect) holds merchant prices and acceptance fixed while networks re-optimize fees and rewards. Consumers gain from reduced credit aversion but lose from lower rewards. The second row (Price Passthrough) allows merchants to adjust retail prices, which generates large consumer gains as fee reductions flow through to lower prices. The third row (Merchant Adoption) allows merchants to adjust acceptance, with a small effect. The decomposition shows that the credit-aversion costs consumers avoid are the main source of welfare gains, while changes in merchant acceptance play a comparatively smaller role.

> Allowing for substitution between payment types may increase the role of new payment methods (R2 point 2d). Depending on your ability to adopt the above suggestions, you might be able to engage more with this or at a minimum note that demand limits the impact of new payment methods.

Reply: I have added a discussion when introducing the public debit counterfactual in Section VI.E of how the substitution assumption limits the effect of new payment types. The key idea is that if consumers are unwilling to substitute between the new payment method and existing ones at the point of sale, then it is unlikely to generate downward pressure on merchant fees. While this may seem mechanical, it is consistent with the fact that low fees on debit cards have not generated any downward pressure on credit card merchant fees.

> I also like R1's suggestion to investigate the importance of consumer multihoming, which would speak to the broader two- sided markets literature. That said, if you find that this takes you too far afield from presenting a coherent story with the counterfactuals, I would also be fine with leaving this point for future work

Reply: I have added a "Dual Routing" counterfactual in Section VI.D that increases the share of multi-homing consumers by 20 pp, matching policy proposals such as the Credit Card Competition Act. Consistent with the theoretical predictions of Teh et al. (2023), increased multihoming leads networks to compete more intensely for merchant acceptance. Credit fees fall by 13 bp and rewards decline by 15 bp. Total welfare improves by \$3.8 billion (net of the mechanical χ^{22} adjustment described in Online Appendix OE.7).

> Lastly, R4 asks about unregulated optimal behavior. My guess was that this serves as the baseline for the counterfactuals already, but please correct me if I am wrong.

Reply: The baseline is the status quo equilibrium in which debit cards are regulated but credit cards are not. Each counterfactual then perturbs this baseline. For example, "Cap Fees" imposes interchange caps, while "Uncap Debit" removes the existing Durbin constraint. I have clarified this in the counterfactuals section.

> "My main concern while reading the paper relates to the exposition. The writing is reasonably well structured and polished, but I still find it hard to read. I think this is partly because the author sometimes takes for granted that a reader knows the institutions. For example, the first page refers to "poorly designed price regulations" that imply distortions but is vague on what these regulations mean. The second page then describes the finding on the impact of the basis point reduction in debit rewards after the Durbin Amendment Act, but without much context. This Act is further referred to a few times in the introduction but only explained at a late stage. In general, the author could do a better job in presenting the paper at a higher level, for a broader audience that is not necessarily familiar with the regulations."
>
> Please take a stab at further tightening the paper's exposition with the objective of (a) clarifying what can and cannot be done with the data at hand, and (b) focusing the paper on the essential pieces of the setting that are relevant for the modeling and the interpretation of your results. I realize this was your job market paper, and it still essentially reads as one, with a lot of detail on nuances of the payments market that are unlikely to be material to your results. You may also want to shorten the footnotes and shorten the Appendix, which contains further details that – while interesting – are not central to the substance of the paper (for example, the detailed section on price coherence). To the extent that a reader would like to engage with the Appendix material, I would recommend that you include only the most important material related to the data compilation and collection, the empirical estimation approach, and the model derivation.

Reply: I have substantially streamlined the paper. The main changes are:

1. *Introduction.* I have rewritten the introduction to be accessible to readers unfamiliar with payment market institutions. The Durbin Amendment and its institutional context are now introduced in the opening paragraphs rather than being referenced early and explained late, and the framing emphasizes standard IO intuitions (market power, competitive bottlenecks, price coherence) rather than payment-specific jargon.

2. *Appendix reorganization.* The main appendices (A–C) now cover the three categories the associate editor recommended: data compilation and collection, model derivation, and the empirical estimation approach. I have moved the detailed section on price coherence, the merchant sorting analysis, and other supplementary material to a separate Online Appendix (OA–OH).

3. *Footnotes and institutional detail.* I have shortened footnotes throughout and removed extended discussions of payment market nuances that are not material to the results.

> Lastly, Frank Verboven suggests connecting with Knittel and Stango's paper "Price Ceilings as Focal Points for Tacit Collusion" (AER 2003) on price ceilings as focal points for collusion in credit card markets. Since this relates closely to price caps, it may be useful to discuss whether institutions have changed since their analysis and whether caps may create a risk of collusion in your setting (not at all to be dealt with, but to mention as a caveat)

Reply: Knittel and Stango (2003) show that state-level non-binding interest rate ceilings served as focal points for tacit collusion in credit card markets during the 1980s. I have added a caveat to the literature review noting that the mechanism is limited in the current setting. The caps I study are binding constraints, whereas the Knittel and Stango result applies to non-binding ceilings that serve as coordination devices above the competitive price.

References

Knittel, Christopher R., and Victor Stango. 2003. “Price Ceilings as Focal Points for Tacit Collusion: Evidence from Credit Cards.” *American Economic Review* 93, no. 5 (December): 1703–1729. Accessed February 1, 2026.

Teh, Tat-How, Chunchun Liu, Julian Wright, and Junjie Zhou. 2023. “Multihoming and Oligopolistic Platform Competition.” *American Economic Journal: Microeconomics* 15 (4): 68–113.